Sri Chaitanya ACADEMY | Infinity Learn

Test Date: 13th Apr 2025



JEE FINAL TEST SERIES

JEE Advanced Part Test - 2_P2 [2023 P1]

Grade XII Full Syllabus

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCE-2023-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 3)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 4 – 7)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 8 – 13)	Questions with **Non-Negative Integer** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 20)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 21 – 24)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 25 – 30)	Questions with **Non-Negative Integer** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 37)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – II(Q.N : 38 – 41)	Questions with Single Correct Choice	+3	-1	4	12
Sec – III(Q.N : 42 – 47)	Questions with **Non-Negative Integer** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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01. A line is drawn through the point $P(-1,1)$ to meet the curve $y=\frac{1}{x}$ in the points A and B (Points A and B lie on same side of P). A point R is chosen on this line such that PA, PR and PB are in Arithmetic Progression. Then the locus of R may be

A) $2xy=1$ **B)** $2xy=x-y$ **C)** $2xy=x+y$ **D)** $x-y=2$

02. For $P(A)=\frac{3}{8};P(B)=\frac{1}{2};P(A\cup B)=\frac{5}{8}$ which of the following do/does hold good?

A) $P\left(A^c/B\right)=3P\left(A/B^c\right)$ **B)** $P(B)=P\left(A/B\right)$

C) $15P\left(A^c/B^c\right)=8P\left(B/A^c\right)$ **D)** $P\left(A/B^c\right)=P(A\cap B)$

3. Given a relation $S:\left\{(x,y):x^2+y^2\le 2024, x,y\in R\right\}$. Which of the following is/are INCORRECT?

A) S is reflexive **B)** S is symmetric
C) S is transitive **D)** S is anti-symmetric

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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4. If $A=\begin{bmatrix}1+x^2-y^2-z^2 & 2(xy+z) & 2(zx-y)\\ 2(xy-z) & 1+y^2-z^2-x^2 & 2(yz+x)\\ 2(zx+y) & 2(yz-x) & 1+z^2-x^2-y^2\end{bmatrix}$, then $|A|$ is equal to

A) $(1+xy+yx+zx)^3$ **B)** $\left(1+x^2+y^2+z^2\right)^3$

C) $(xy+yx+zx)^3$ **D)** $\left(1+x^3+y^3+z^3\right)^2$

5. Two raw data sets of size 10 each are given in two variables x_i and y_i, $i=2024$ to $i=2033$ satisfying $\sum_{i=2024}^{2033} x_i = 235, \sum_{i=2024}^{2033} y_i = 250, \sum_{i=2024}^{2033} x_i^2 = 6750, \sum_{i=2024}^{2033} y_i^2 = 6840$. The standard deviation of the data sets will be

A) 11.08,7.68 **B)** 11.02,7.58 **C)** 11.48,7.48 **D)** 11.08,7.58

6. Let f & g be two functions both being defined from $R \to R$ as follows: $f(x)=\frac{x+3|x|}{2}$ and $g(x)=\begin{cases} x, & x<0 \\ x^2, & x\geq 0 \end{cases}$, then which of the following is/are CORRECT?

A) fog is continuous & differentiable everywhere

B) fog is discontinuous

C) gof is continuous but not differentiable

D) gof is continuous& differentiable everywhere

7. For any triangle ABC, if a,b,c are the lengths of the sides of the triangle, then the expression $(b+c-a)(c+a-b)(a+b-c)-abc$ is

A) Positive **B)** Negative **C)** Non-positive **D)** Non-negative

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

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8. Let $f(n)=\int_0^{\pi/2}\cos^n x\cos nx\,dx$, where n is an odd positive integer. Then the value of $\frac{f(2023)}{f(2024)}$ is ___

9. If the four planes $x+y=0, y+z=0, z+x=0$ & $x+y+z=6$ represent the faces of a Tetrahedron, then the integer near to area of base (in sq. units) of the tetrahedron (on the plane $x+y+z=6$) is ___

10. If a,b,c be three distinct natural numbers in Arithmetic Progression and $a+b+c=2025$, then the possible number of ordered pairs (a,b,c) is ___

11. If the coefficient of $x^r\left(0 \le r \le n-1\right)$ in the expression $\left(x+2\right)^{n-1}+\left(x+2\right)^{n-2}\left(x+1\right)+\left(x+2\right)^{n-3}\left(x+1\right)^2+.....+\left(x+1\right)^{n-1}$ is denoted by ${}^n_r b$, then ${}^{10}_7 b$ is equal to

12. Let a,b,c,d be distinct integers such that the equation $\left(x-a\right)\left(x-b\right)\left(x-c\right)\left(x-d\right)=9$ has an integer root 'r', then the value of $a+b+c+d-4r$ is equal to

13. Given positive rational numbers a,b,c such that $a+b+c=1,$ then the maximum value of $a^a b^b c^c + a^b b^c c^a + a^c b^a c^b$ is

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14.

	List I		List II
(P)	If $2\sin x, \sin 2x$ & $2\cos x$ are in Arithmetic Progression, then the value of $\left\|\sin x+\cos x\right\|$ is	(1)	$\frac{\sqrt{5}-1}{2}$
(Q)	If the angles of a triangle are in Arithmetic Progression with the common difference equal to $\frac{1}{3}$ of the greatest angle, then ratio of the greatest side to second greater side is	(2)	$\frac{2}{\sqrt{3}}$
(R)	Let the side AB, BC, CD & DA of a cyclic quadrilateral ABCD are in Geometric Progression with common ratio $\sqrt{3}$, then the value of $\frac{BD}{AB}$ is	(3)	$2\sqrt{5}$
(S)	Let a,b,c be in Arithmetic Progression& a^2,b^2,c^2 be in Harmonic Progression. If $a \ne c$, then $a:c$ is	(4)	$\sqrt{3}-2$

A) (P) – (1), (Q) – (4), (R) – (3), (S) – (2)

B) (P) – (3), (Q) – (2), (R) – (1), (S) – (4)

C) (P) – (1), (Q) – (2), (R) – (3), (S) – (4)

D) (P) – (3), (Q) – (4), (R) – (1), (S) – (2)

15.

List-I		List-II	
(P)	If the straight lines $y = a_1x + b_1$ and $y = a_2x + b_2$, $(a_1 \neq a_2)$ and $b_1 \neq b_2$ meet the coordinate axes in concyclic points, then	(1)	$a_1^2 + a_2^2 = 1$
(Q)	If the chord of contact of the tangents drawn to $x^2 + y^2 = 4$ from any point on $x^2 + y^2 = a_1^2$ touches the circle $x^2 + y^2 = a_2^2, (a_1 \neq a_2), (a_1, a_2 > 0)$, then	(2)	$a_1 + a_2 = 20$
(R)	If the circles $x^2 + y^2 + 2a_1x + 2a_2y - 1 = 0$ and $x^2 + y^2 - 2a_1x - 2a_2y - 1 = 0$, $(a_1 \neq a_2)$ cut orthogonally, then	(3)	$a_1a_2 = 4$
(S)	If the shortest and largest distance from the point $(8,5)$ to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ are a_1 and a_2 respectively, then	(4)	$a_1a_2 = 1$

A) (P) – (4), (Q) – (3), (R) – (1), (S) – (2)
B) (P) – (3), (Q) – (4), (R) – (1), (S) – (2)
C) (P) – (4), (Q) – (3), (R) – (2), (S) – (1)
D) (P) – (3), (Q) – (4), (R) – (2), (S) – (1)

16.

List-I		List-II	
(P)	$P(A \mid C) + P(B \mid C) - P((A \cap B) \mid C)$	(1)	$P(\overline{A} \mid (\overline{B} \cap \overline{C}))$
(Q)	$P((A \cap \overline{B}) \mid C) + P((A \cap B) \mid C) =$	(2)	$P(A \mid C)$
(R)	$\dfrac{1 - P(A) - P(B \cup C) + P(A \cap (B \cup C))}{1 - P(B \cup C)} =$	(3)	$P((A \cup B) \mid C)$
(S)	$P(\overline{A} \cup \overline{B}) + P(A \cap B)$	(4)	1

A) (P) – (4), (Q) – (3), (R) – (1), (S) – (2)
B) (P) – (3), (Q) – (2), (R) – (1), (S) – (4)
C) (P) – (4), (Q) – (3), (R) – (2), (S) – (1)
D) (P) – (3), (Q) – (4), (R) – (2), (S) – (1)

17. In a ΔAEX ,T is the mid-point of XE, and P is the mid-point of ET. If the ΔAPE is equilateral of side length equal to unity

List-I		**List-II**	
(P)	$\frac{1}{AX} =$	(1)	1
(Q)	$\cos\left(\angle XAE\right) =$	(2)	$-\frac{1}{\sqrt{13}}$
(R)	$\sin\left(\angle EAT\right) =$	(3)	$\frac{1}{\sqrt{13}}$
(S)	$\cot\left(\angle APX\right) =$	(4)	$-\frac{1}{\sqrt{3}}$

A) (P) – (4), (Q) – (3), (R) – (1), (S) – (2)

B) (P) – (3), (Q) – (3), (R) – (1), (S) – (2)

C) (P) – (3), (Q) – (2), (R) – (1), (S) – (4)

D) (P) – (3), (Q) – (4), (R) – (2), (S) – (1)

PHYSICS **Max Marks: 60**

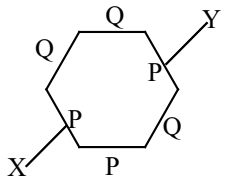
SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

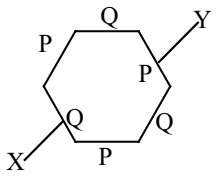
This section contains **THERE (03)** questions.

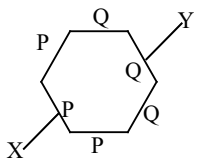
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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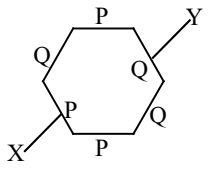
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
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For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

18. In the following hexagons, made up of two different material P and Q, current enters and leaves from points X and Y respectively. In which case, the magnetic field at its centre is not zero:

A) 

B) 

C) 

D) 

19. The ground state and first excited state energies of hydrogen atom are $-13.6eV$ and -3.4 eV respectively. If potential energy in ground state is taken to be zero, then

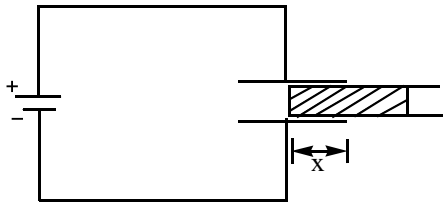
A) Potential energy in the first excited state would be 20.4 eV

B) Total energy in the first excited state would be 23.8 eV

C) Kinetic energy in the first excited state would be 3.4 eV

D) Total energy in the ground state would be 13.6 eV.

20. A capacitor is charged by using a battery and then a dielectric slab is released from the position shown in the figure (battery remains connected). Then, (Dielectric has the same dimensions as the space between plates and there is no friction, x is the length of the part of dielectric inserted into the capacitor)



A) Force on dielectric slab does not depend on the dimension x

B) Force on dielectric slab depends on the dimension x

C) Time period of oscillation does not depend on the dimension x

D) Time period of oscillation depends on the dimension x

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

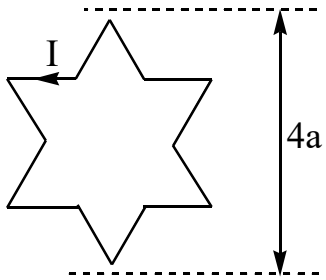
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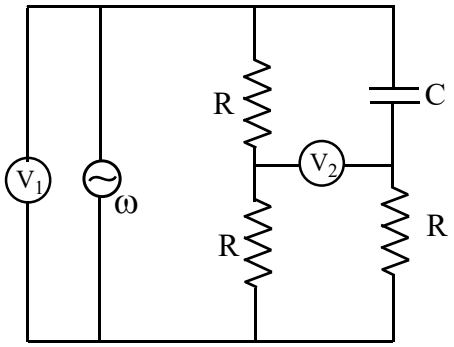
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21. A symmetric star conducting wire loop is carrying a steady state current I as shown figure. The distance between the diametrically opposite vertices of the state is 4a. The magnitude of the magnetic field at the centre of the loop is

A) $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}-1\right]$ **B)** $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}+1\right]$ **C)** $\frac{\mu_0 I}{4\pi a}3\left[\sqrt{3}-1\right]$ **D)** $\frac{\mu_0 I}{4\pi a}3\left[2-\sqrt{3}\right]$

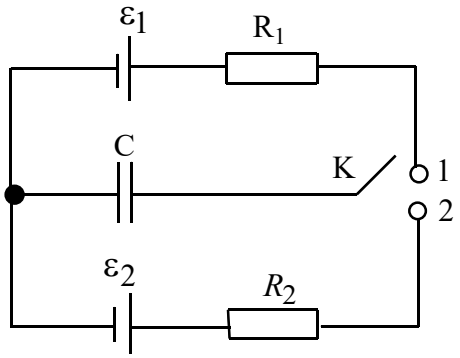


22. A circuit shown consists of three resistors, one capacitor, 2 ideal AC voltmeters and alternating voltage source of angular frequency ω. Take $R\omega C = 1$. If the reading of V_1 is V, then the reading of V_2 is



A) $\frac{\sqrt{5}V}{2}$ **B)** $\frac{V}{2}$ **C)** $\frac{V}{3}\sqrt{\frac{7}{2}}$ **D)** $\frac{V}{3}\sqrt{\frac{5}{2}}$

23. The key 'K' is connected in turn to each of the contacts over short identical time intervals so that the change in the charge on the capacitor over each connection is small. We assume that an almost steady state of the circuit is achieved, when the voltage across the capacitor practically does not change. After that, the charge of the capacitor no longer changes after each switching cycle. Find the steady state charge of the capacitor.



A) $\frac{C(\varepsilon_1 R_2 + \varepsilon_2 R_1)}{R_1 + R_2}$ **B)** $\frac{C(\varepsilon_1 R_2 + \varepsilon_2 R_1)^2}{(R_1 + R_2)^2}$ **C)** $\frac{C(R_2 R_1)}{R_1 + R_2}$ **D)** $\frac{C\varepsilon_1\varepsilon_2(R_1 + R_2)}{(\varepsilon_1 R_2 + \varepsilon_2 R_1)}$

24. The box of a pin hole camera, of length L, has hole of radius a. it is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ, the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size $(\text{say}\, b_{\min})$ when:

A) $a=\sqrt{\lambda L}$ *and* $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$ **B)** $a=\sqrt{\lambda L}$ *and* $b_{\min} = \sqrt{4\lambda L}$

C) $a=\frac{\lambda^2}{L}$ *and* $b_{\min} = \sqrt{4\lambda L}$ **D)** $a=\frac{\lambda^2}{L}$ *and* $b_{\min} = \left(\frac{2\lambda^2}{L}\right)$

SECTION-III

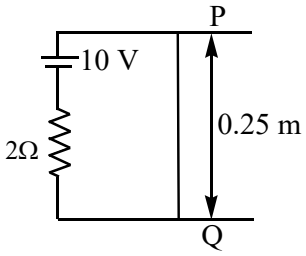
(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

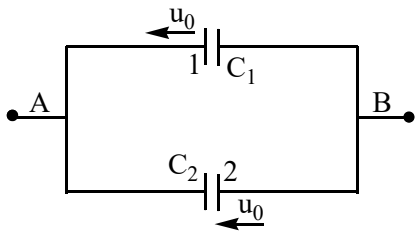
Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. A metal wire PQ slides on parallel metallic rails having separation 0.25 m, each having negligible resistance. There is a 2Ω resistor and 10 V battery as shown in figure. There is a uniform magnetic field directed into the plane of the paper of magnitude 0.5 T. A force of 0.5 N to the left is required to keep the slide wire PQ moving with constant speed to the right. If the speed of the wire PQ is '2x' m/s, then x is _____

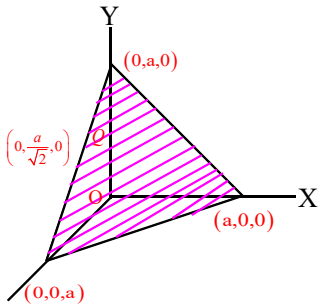


26. Find the quantum number 'n' corresponding to excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelength $1026.7\,\overset{0}{A}$ and $304\,\overset{0}{A}$. $\left(R = 1.096 \times 10^7 / m\right)$

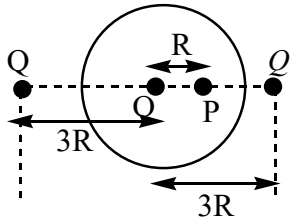
27. Two identical capacitor having plate separation 'd_0' are connected parallel to each other across points A and B as shown in figure. A total charge Q is imparted to the system by connecting a battery across A and B and battery is removed. Now first plate of first capacitor and second plate of second capacitor starts moving with constant velocity u_0 towards left. The magnitude of current flowing in the loop during this process at the initial instant is $\frac{Qu_0}{nd_0}$, find the value of 'n', taking all quantities are in S.I units



28. A point charge Q is placed on y-axis at $\left(0,\frac{a}{\sqrt{2}},0\right)$. Consider a triangular surface having vertices at $(a,0,0),(0,a,0)\,\&\,(0,0,a)$. If electric flux through this triangular surface is $\frac{5Q}{(10a+b)\varepsilon_0}$, where a, b are integers, find $(a+b)$.



29. A solid conducting sphere of center 'O', radius 2R, carrying charge Q is surrounded by two point charges Q and Q as shown in the figure. The electric field at point P due to the charges on conducting sphere is $\frac{x}{16}\frac{Kq}{R^2}$, then x is ___ $\left(OP=R \text{ and } K=\frac{1}{4\pi\in_0}\right)$



30. A nucleus with mass number 220 initially at rest emits an α-particle. If the Q-value of the reaction is 5.5 MeV, the kinetic energy of the α-particle is $1.8n\ MeV$, when n is ___

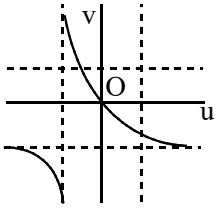
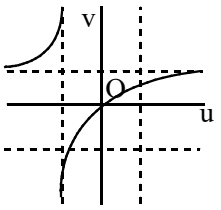
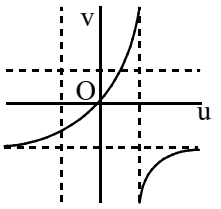
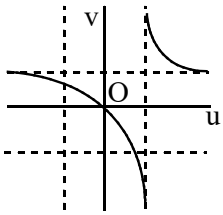
SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

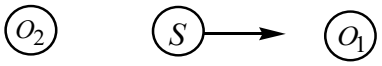
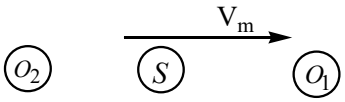
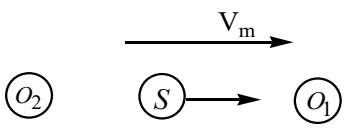
31. Match the following graphs (in column-II) between position of object (u) and position of image (v). Take positive sign in the direction of instant ray.

	Column-I Device		**Column-II Probable Image**
A)	Concave mirror	p)	
B)	Convex mirror	q)	
C)	Convex lens	r)	
D)	Concave lens	s)	

A) $A-p, B-q, C-s, D-r$ **B)** $A-r, B-s, C-q, D-p$

C) $A-p, B-s, C-q, D-r$ **D)** $A-q, B-s, C-p, D-r$

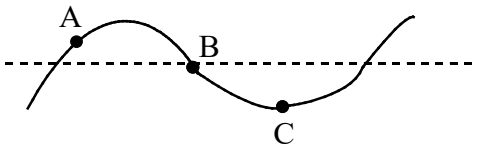
32. A source of sound when stationary with respect to medium, emits sound of frequency f and wavelength λ. Speed of sound with respect to medium is c. Speed of medium is V_m {in case (c) and case (d) of column I}. The observer O_1 receives waves of frequency f_1 and wavelength λ_1. The observer O_2 receives waves of frequency f_2 and wavelength λ_2. Speed of source with respect to ground is v_s {in case (d) of column I}

Column-I		Column-II	
A)	Medium at rest, source at rest, O_1 and O_2 moves as shown. 	P)	$\lambda_1 < \lambda < \lambda_2$
B)	Medium at rest, O_1, O_2 at rest, source moves as shown. 	Q)	$f_1 < f < f_2$
C)	Medium moves as shown, source, O_1 and O_2 are at rest. 	R)	$\lambda_1 = \lambda = \lambda_2$
D)	Medium moves, source moves, O_1 and O_2 are at rest. $(V_s > V_m)$ 	S)	$\lambda_1 > \lambda > \lambda_2$

A) $A-R; B-S; C-P,Q; D-S$ **B)** $A-R; B-P,Q; C-S; D-P$

C) $A-S; B-P,Q; C-S; D-S$ **D)** $A-R; B-P,Q; C-S; D-P,Q$

33. The figure shows a string at a certain moment as a transverse wave passes through it. Three particles A, B and C of the string are also shown. Point A is in the motion while B is at mean position and point C is at the extreme. Match the physical quantities in the left column with the description in the column on the right

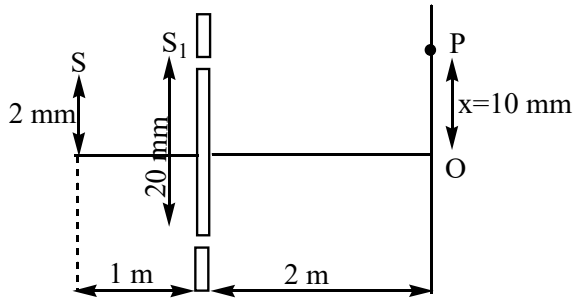


Column-I		Column-II	
A)	Velocity of A	P)	Downwards, if the wave is travelling towards right.
B)	Acceleration of A	Q)	Downwards, if the wave is travelling towards left
C)	Velocity of B	R)	Downwards, no matter which way the wave is travelling
D)	Velocity of C	S)	Zero.

A) $A-Q; B-Q; C-R; D-S$ **B)** $A-P; B-Q; C-R; D-R$

C) $A-P; B-R; C-Q; D-S$ **D)** $A-Q; B-R; C-Q; D-R$

34. In young's double-slit experiment, the point source S is placed slightly off the central axis as shown in figure. If $\lambda = 500nm$, then match the following.



Column-I		Column-II	
A)	Nature and order of interference of the point P, OP=10 mm	p)	Bright fringe of order 80
B)	Nature and order of interference at point O	q)	Bright fringe of order 262
C)	If a transparent paper (refractive index $\mu = 1.45$) of thickness t=0.02 mm is pasted on S_1, i.e., one of the slits, the nature and order of the interference at P	r)	Bright fringe of order 62
D)	After inserting the transparent paper (as given in C) front of slit S_1, the nature and order of interference at O	s)	Bright fringe of order 280

A) $A-s, B-p, C-q, D-r$ **B)** $A-p, B-r, C-s, D-q$

C) $A-q, B-r, C-p, D-r$ **D)** $A-q, B-s, C-r, D-p$

CHEMISTRY **Max Marks: 60**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

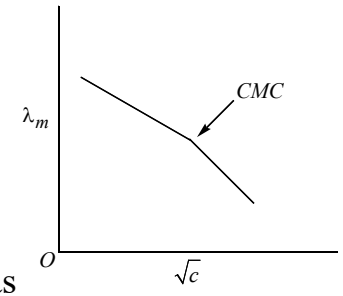
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

35. Which of the following statements are correct?

A) The cleansing action of soaps involves encapsulation of dirt by 'R' part of $RCOO^-$

B) Oxidation of oxalic acid by $KMnO_4$ in acidic medium is an example of autocatalysis

C) Variation of molar conductivity (λ_m) with concentration of sodium stearate in solution can be represented as



D) The coagulating power of electrolytes having ions $Na^{\oplus}, Al^{3+}$ *and* Ba^{2+} for arsenic sulphide solution increases in the order. $Na^{\oplus} < Ba^{2+} < Al^{3+}$

36. Gluconic acid is not obtained in

A) $Glucose \xrightarrow{Br_2/H_2O}$ **B)** $Glucose \xrightarrow[(ii)H_3O^+]{(i)Tollen's\ \text{Reagent}}$

C) $Glucose \xrightarrow[(ii)H_3O^+]{(i)Fehling's\ solution}$ **D)** $Glucose \xrightarrow{Conc.\ HNO_3}$

37. Which of the following Statement(s) is/are correct?

A) In group-18, radon is obtained as a decay product of ^{226}Ra

B) Noble gases have positive values of electron gain enthalpy.

C) $\Delta S < 0$, for $\left[Ni(NH_3)_6\right]^{2+} + 3en \rightarrow \left[Ni(en)_3\right]^{2+} + 6NH_3$

(Where en=ethylene diamine)

D) A mixture of three parts of concentrated HCl and one part of concentrated HNO_3 is used for dissolving gold, platinum.

SECTION – II
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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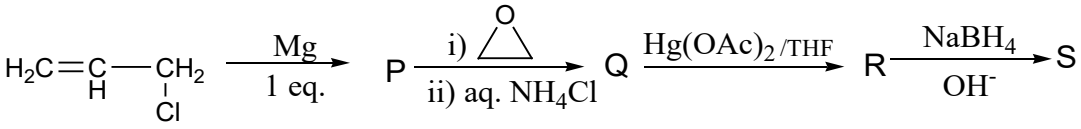
38. The E_{cell} for a given cell is 1.2346V and 1.2340V at 300K and 310K respectively, calculate ΔS for the cell reaction if the redox change involves three electrons. E_{cell} varies linearly with temperature.

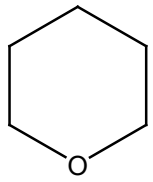
A) -17.37 J/k **B)** -13.58J/k **C)** -12.65J/k **D)** -8.76J/k

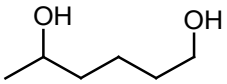
39. A compound has the empirical formula $C_{10}H_8Fe$. A solution of 0.26 g of the compound in 11.2 g of benzene (C_6H_6) boils at $80.26^\circ C$. The boiling point of benzene is $80.10^\circ C$; the K_b of benzene is $2.53^\circ C / molal$. What is the molecular formula of the compound? (Atomic mass of Fe: 56)

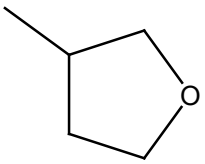
A) $C_{30}H_{24}Fe_3$ **B)** $C_{10}H_8Fe$ **C)** C_5H_4Fe **D)** $C_{20}H_{16}Fe_2$

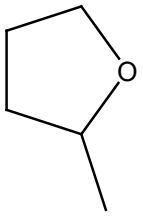
40. The major product(s) obtained in the following reactions is



A)


B)


C)


D)


41. Which of the following statements is incorrect?

A) Wrought iron (or malleable iron) is the purest form of commercial iron and is prepared from cast iron by oxidizing impurities in a reverberatory furnace lined with haematite.

B) 'Copper matte' is a mixture of Cu_2S and FeS.

C) From low grade copper ores, copper is extracted by hydrometallurgy (acid or bacteria leaching followed by reduction with scrap iron or H_2).

D) The reduction of a metal oxide is very difficult if the metal formed is in liquid state at the temperature of reduction.

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

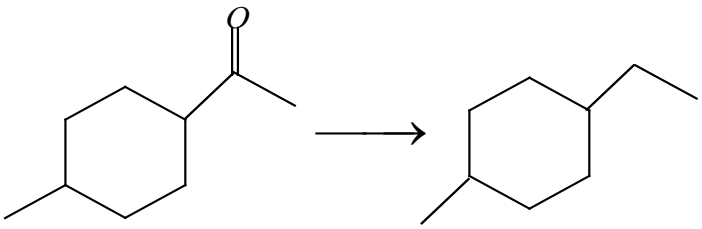
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42. The K_{sp} of AgI is calculated as 8×10^{-y} where 'y' is an integer. If $y=(2x+1)$ then what is the value of 'x'. Given that $E^{\circ}_{Ag^{+}/Ag}=0.799V$ $\quad$ $E^{\circ}_{I^{-}/AgI/Ag}=-0.151V$ *at* $25^{\circ}C$

43. AgX crystallizes in rock salt like structure in which X^{-} occupy FCC lattice site. The density of AgX crystal found to be $\frac{40}{6.022}g/cm^{3}$ and the edge length of the unit cell is 600 pm. What will be the mass(in grams) of 1 mole unit cell if all the atoms along one body diagonal of each unit cell have been removed. [Given: Atomic weight of Ag = 108 g/mol, $N_A=6.022\times10^{23}$]

44. For the complete reduction of 58g of acetone to isopropyl alcohol, the quantity of $LiAlH_4$ required (assuming chemical yield to be 100%) is approximately
[Mass: Li = 7, Al = 27]

45. How many reagent(s) will be appropriate for the following reaction?



(Major product)

I) Red P + HI $\quad$ II) LAH $\quad$ III) N_2H_4/H_2O_2 $\quad$ IV) $N_2H_4/OH^{-},\Delta$

V) HS $\overset{SH}{\smile}$, followed by H_2/Ni $\quad$ VI) $Na/EtOH$ $\quad$ VII) DIBAL – H

VIII) $NaBH_4$ $\quad$ IX) H_2/Ni $\quad$ X) $Zn-Hg/HCl$

46. Find the total number of electron of $Co^{2+}ion$ which are present in non-axial d-orbitals in the complex $\left[Co(dmg)_2\right]$.

47. Acetate ion gives deep red colour on reaction with neutral ferric chloride solution due to the formation of complex ion(x) which decomposes on heating to give a brown red precipitate(y). Then the total number of CH_3COO^{-} groups present in one molecule/ion each of x and y is_____.

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

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48. If t_x represents time required for a fraction x of a reactant to be reacted during a reaction involving only one reactant. Then match List-I with List-II.

List-I		List-II	
(I)	Zero order reaction	P)	$\frac{t_{3/4}}{t_{1/2}} = 3$
(II)	First order reaction	Q)	$\frac{t_{1/4}}{t_{1/8}} = 2$
(III)	Second order reaction	R)	$\frac{t_{7/8}}{t_{3/8}} = \frac{3}{2}$
(IV)	Order of reaction ≥ 1	S)	$\alpha = \left(1 - e^{-kt}\right)$
		T)	$\frac{t_{100\%}}{t_{50\%}} = 2$

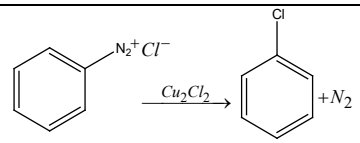
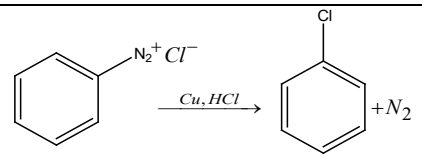
A) $I \to Q,T; II \to R,S; III \to P; IV \to P,R,S$

B) $I \to Q; II \to R,T; III \to P; IV \to P,R$

C) $I \to T; II \to P; III \to R,S; IV \to Q,T$

D) $I \to R; II \to Q; III \to P; IV \to P,Q$

49. Match List-I with List-II

List-I		List-II	
a)		i)	Wurtz reaction
b)		ii)	Sandmeyer reaction
c)	$2CH_3CH_2Cl + 2Na \xrightarrow{Ether} C_2H_5 - C_2H_5 + 2NaCl$	iii)	Fitting reaction
d)	$2C_6H_5Cl + 2Na \xrightarrow{Ether} C_6H_5 - C_6H_5 + 2NaCl$	iv)	Gatterman reaction

Choose the correct answer from the options given below:

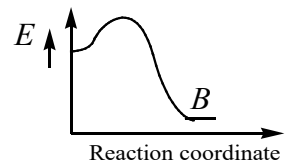
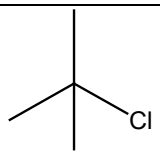
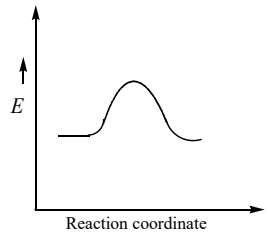
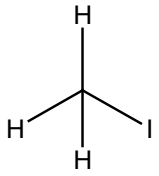
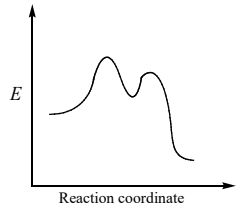
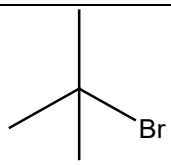
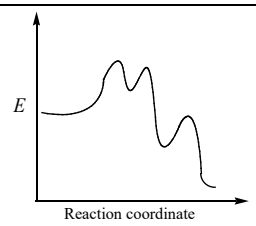
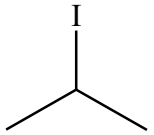
A) $(a) \to (iii), (b) \to (i), (c) \to (iv), (d) \to (ii)$

B) $(a) \to (ii), (b) \to (i), (c) \to (iv), (d) \to (iii)$

C) $(a) \to (iii), (b) \to (iv), (c) \to (i), (d) \to (ii)$

D) $(a) \to (ii), (b) \to (iv), (c) \to (i), (d) \to (iii)$

50. Match each of the following transformation to the correct reaction properties

Column-I		Column-II	
P)		1)	
Q)		2)	
R)		3)	
S)		4)	

A) $P-4, Q-2, R-1, S-3$ **B)** $P-1, Q-2, R-3, S-4$

C) $P-4, Q-2, R-none, S-3$ **D)** $P-4, Q-2, R-3, S-1$

51.

List-I (Oxoacids of sulphur)	List-II (Characteristic bonds)
P. Pyrosulphuric acid $(H_2S_2O_7)$	1. $S-S$ bond (single bond)
Q. Dithionic acid $(H_2S_2O_6)$	2. $S=S$ bond(double bond)
R. Thiosulphuric acid $(H_2S_2O_3)$	3. $S-O-S$ bond
S. Pyrosulpurous acid $(H_2S_2O_5)$	4. $S=O$ bond(double bond)
	5. $S-OH$ bond

Correct match is:

A) P-3, 4, 5; Q-1, 4, 5; R-2, 4, 5; S-1, 4, 5

B) P-1, 3, 4; Q-1, 4, 5; R-2, 4, 5; S-3, 4, 5

C) P-3, 4, 5; Q-3, 4, 5; R-2, 4, 5; S-2, 4

D) P-3, 4; Q-2, 5; R-3, 5; S-2, 5